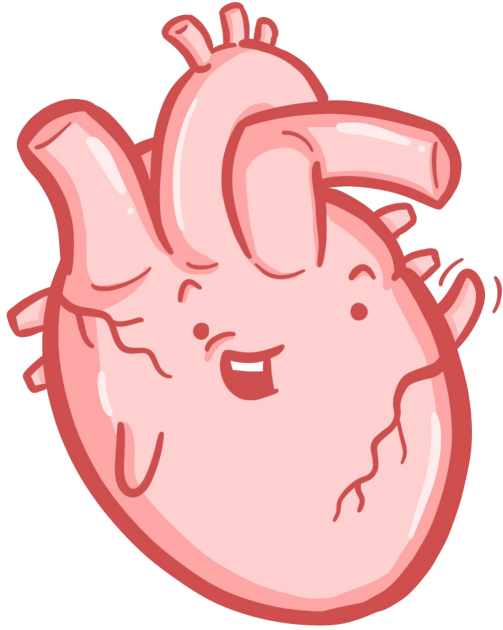




Major arteries Of The body

Editing file

Cardiovascular block-Anatomy-Lecture 3



Objectives

Color guide :

Only in boys slides in **Green**

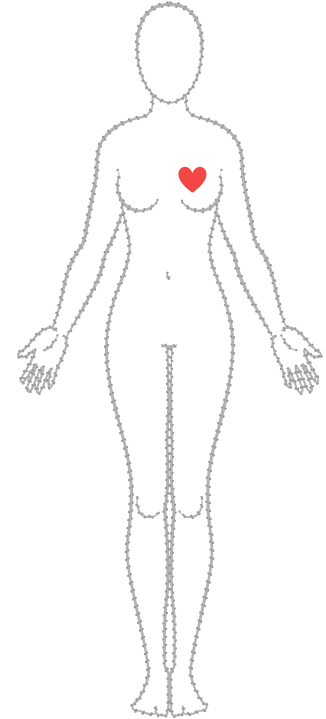
Only in girls slides in **Purple**

important in **Red**

Notes in **Grey**

At the end of the lecture, the student should be able to:

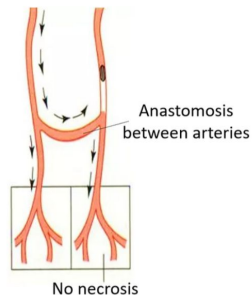
- Define the word 'artery' and understand the general principles of the arterial system.
- Define arterial anastomosis and describe its significance.
- Define end arteries and give examples.
- Describe the aorta and its divisions & list the branches from each part.
- List major arteries and their distribution in the head & neck, thorax, abdomen and upper & lower extremities.
- List main pulse points.



Arteries and their general principles

- Arteries are blood vessels that carry blood from the heart to the body.
- ALL Arteries carry oxygenated blood, **EXCEPT** the **PULMONARY ARTERY**, carrying deoxygenated blood to the lung.
- The flow of blood depends on the pumping action of the heart.
- Arteries have **Elastic Wall** containing **NO valves**.

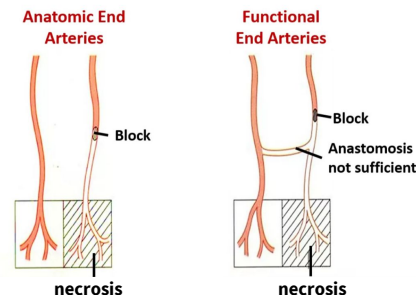
❖ The branches of arteries supplying adjacent areas normally **ANASTOMOSE** with one another freely providing backup routes for blood to flow if one artery is blocked, e.g. arteries of limbs , **GIT**



❖ The arteries whose terminal branches do **NOT ANASTOMOSE** with branches of adjacent arteries are called **"END ARTERIES"**

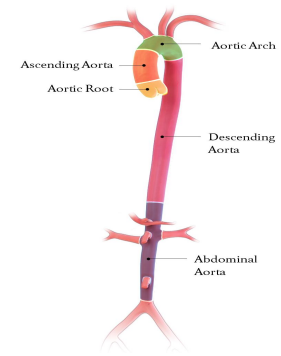
→ it has 2 types:

1. **Anatomic (True) End Artery:** When **NO** anastomosis exists, e.g. artery of the retina. this artery is affected frequently in diabetic patients
2. **Functional End Artery:** When an anastomosis exists but is incapable of providing a sufficient supply of blood, e.g. splenic artery, renal artery.



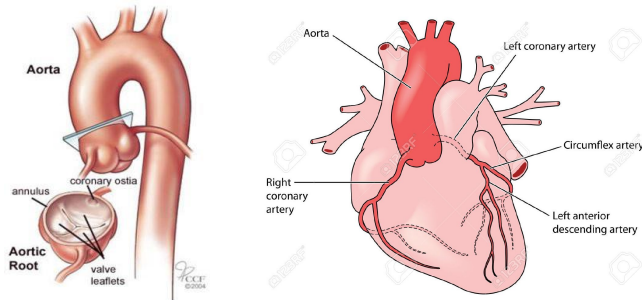
Aorta :

- The **largest** artery in the body .
- Carries oxygenated blood to all part of the body .
- Divided into **4 parts** :
 1. Ascending aorta
 2. Arch of aorta
 3. Descending aorta
 4. Abdominal aorta



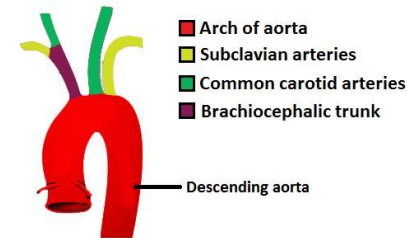
Ascending aorta

- Originated from left ventricle .
- Continues as the **arch of aorta** .
- Has **3** dilatations at its base called **aortic sinus** .
- ❖ Branches :
 1. **Right coronary arteries**
 2. **Left coronary arteries**
 } supplying heart, arise from aortic sinuses



Arch of aorta

- Continuation of the **ascending aorta** . (at the level of T4)
- Leads to descending aorta .
- located **behind the lower part of manubrium sterni** and on the left side of trachea .
- ❖ Branches :
 1. **Brachiocephalic trunk** .
 2. **Left common carotid artery** .
 3. **Left subclavian artery** .



Carotids

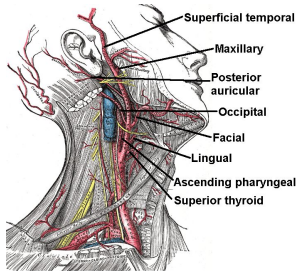
- Origin: **LEFT** from aortic arch. **RIGHT** from brachiocephalic trunk.
- At the level of the disc between C3 & C4, each common carotid divides into two branches:

External carotid

It divides behind neck of mandible into:
Superficial temporal & maxillary arteries

It supplies:

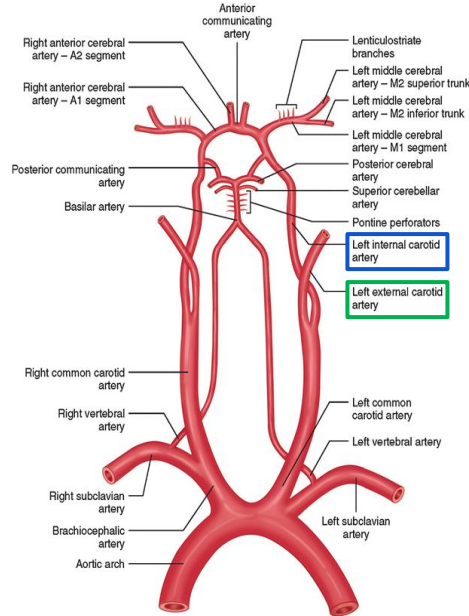
- **Scalp:**
 1. Superficial temporal artery
 2. Occipital artery
 3. posterior auricular arteries
- **Face:** Facial artery
- **Maxilla & mandible:** Maxillary artery
- **Tongue:** Lingual artery
- **Pharynx:** ascending pharyngeal artery
- **Thyroid gland:** Superior thyroid artery



Mnemonic: from bottom to top

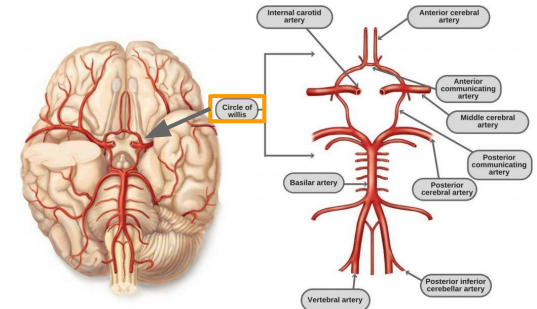
Some American Ladies Found Our
Pants So Magnificent

Some Anatomists Like Freaking
Out Poor Medical Students



Internal carotid

- Has NO branches in the neck
- Enters the cranial cavity, joins the **basilar artery** (formed by the **union** of two vertebral arteries) and forms '**arterial circle of Willis**' to supply **brain**
- In addition, it supplies
 1. Nose
 2. Scalp
 3. Eye

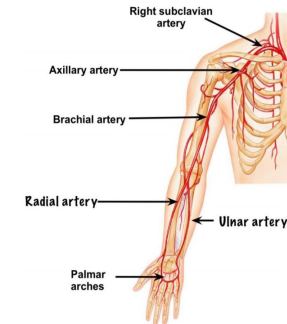
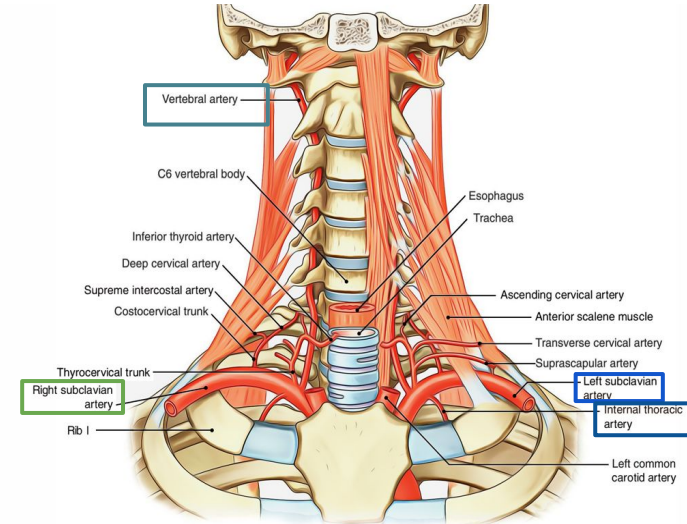


Subclavian Artery & Arteries of the upper limb

- **Origin:**
Left: from arch of aorta.
Right: from brachiocephalic trunk
- **Main branches of subclavian artery :**
 1. **Vertebral artery:** supplies brain and spinal cord
 2. **Internal thoracic artery:** supplies thoracic wall (it's give anterior intercostal arteries)

MSK Flashback 🦋

- The **subclavian artery** continues at lateral border of first rib as:
 - **Axillary artery**
- then at the lower border of teres major it becomes
 - **Brachial artery**
- Until it reaches the end of the cubital fossa, it divides into:
 - **Radial artery**
 - **Ulnar artery**



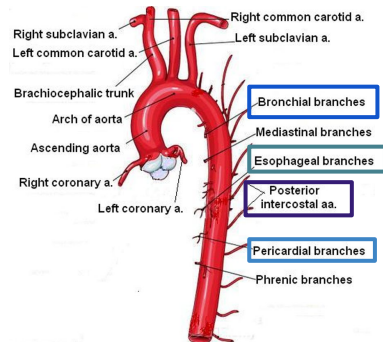
Descending thoracic and Abdominal aorta

→ Descending thoracic aorta

- It is the **continuation of the aortic arch**
- At the level of the **T12**, it passes through the diaphragm and continues as the **abdominal aorta**

• Branches:

- ☐ **Pericardial**
- ☐ **Esophageal**
- ☐ **Bronchial**
- ☐ **Posterior intercostal**



→ Abdominal aorta

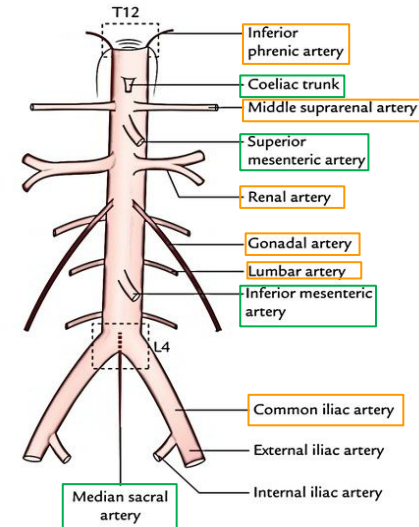
- It enters the abdomen through the **aortic opening of diaphragm**.
- **At the level of lower border of L4, it divides into two common Iliac arteries.**
- **Branches:** divided into two groups:

• Single branches:

1. Coeliac artery
2. Superior mesenteric artery
3. Inferior mesenteric artery
4. Median sacral artery

• Paired branches:

- ☐ inferior phrenic arteries
- ☐ middle suprarenal arteries
- ☐ renal arteries
- ☐ gonadal arteries
- ☐ Four pairs of Lumbar arteries
- ☐ common iliac arteries



Common iliac artery & Arteries of the lower limb

1. **Internal iliac artery** : Supplies the pelvis

2. **External iliac artery** :

→ Continues (behind the midpoint of the inguinal ligament) as :

- **Femoral artery**

☐ It is the main arterial supply of the lower limb

→ Passes through the adductor hiatus and continues as:

- **Popliteal Artery**

☐ Deeply situated in the popliteal fossa.

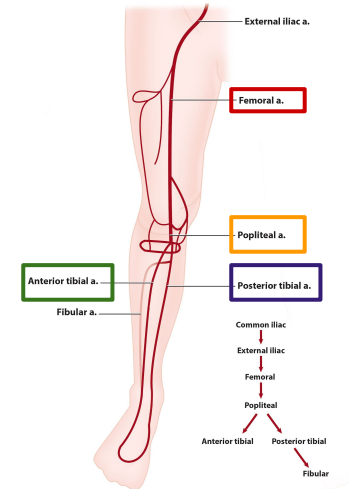
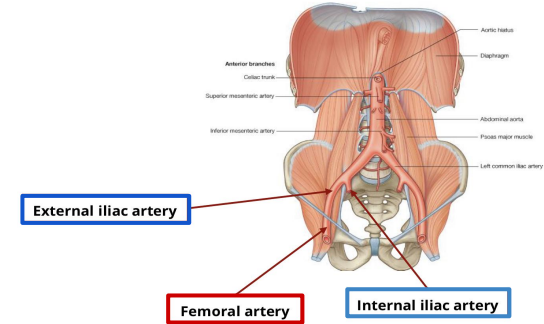
→ Divides, at the lower end of the popliteal fossa into:

1. **Anterior Tibial Artery** continue as:

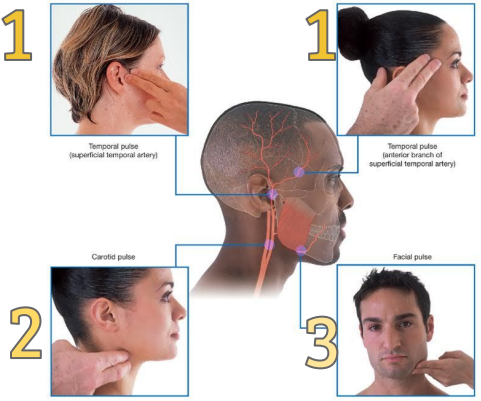
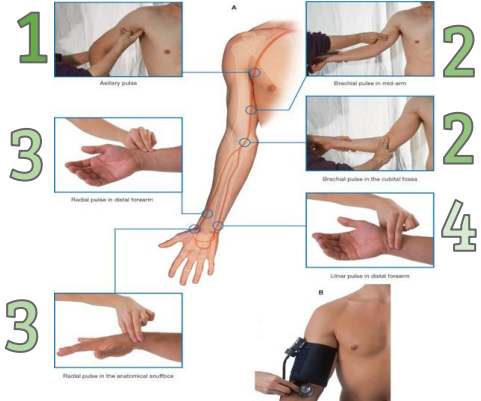
- **Dorsalis pedis artery**

2. **Posterior Tibial Artery** continue as:

- **Medial/Lateral plantar arteries**



Main pulse point

Head and Neck	Upper limb	Lower limb
Temporal pulse - Superficial temporal artery - Anterior branch of Superficial temporal artery Carotid pulse (just below the angle of the mandible) Facial pulse (in front of the masseter muscle)	Axillary pulse (lateral wall of the axilla) Brachial pulse in - mid arm - cubital fossa (medial to the biceps tendon) Radial pulse in : - distal forearm (lateral to FCR) - anatomical snuffbox Ulnar pulse (lateral to FCU in the)	Femoral pulse (just below the midpoint of the inguinal ligament) Popliteal pulse (center of the popliteal fossa) Posterior tibial (Behind and below the medial malleolus) Dorsalis pedis pulse (Between the 1st and 2nd metatarsal bones, lateral to EHL)
		

MCQs

Question 1: The brachiocephalic trunk is a branch of ?

- A. ascending aorta
- B. arch of aorta
- C. descending aorta
- D. common carotid artery

Question 2: Which of the following is a branch of the descending aorta?

- A. Anterior intercostal
- B. superior mesenteric
- C. Pericardial
- D. middle suprarenal

Question 3: Abdominal aorta enters the abdomen and divides into two common iliac arteries at the level of ?

- A. lower border of L5
- B. upper border of L4
- C. upper border of L5
- D. lower border of L4

Question 4: Which of the following is a single branch of the abdominal aorta

- A. Inferior mesenteric
- B. common iliac
- C. popliteal artery
- D. Femoral artery

Question 5: Common iliac artery originates from?

- A. brachiocephalic artery
- B. Arch of aorta
- C. subclavian artery
- D. Abdominal aorta

Question 6: ALL arteries carry oxygenated blood, EXCEPT

- A. Umbilical artery
- B. Pulmonary artery
- C. Coronary artery
- D. Both A and B

Question 7: splenic artery is an example of?

- A. Functional End Artery
- B. True End Artery
- C. Anatomic End Artery
- D. Anastomosis artery

Question 8: Esophageal artery is one of the branches of ?

- A. Left subclavian artery
- B. Ascending aorta
- C. Descending thoracic aorta
- D. Right subclavian artery

Team members

Boys team:

- ★ Faisal Al-Qifari
- Salman Alagla
- ★ Ziyad Al-jofan
- ★ Ali Aldawood
- ★ Khalid Nagshabandi
- ★ Omar Alammari

Team leaders

- ★ Abdulrahman Shadid
- Ateen Almutairi

Girls team :

- Ajeed Al Rashoud
- Taif Alotaibi
- Noura Al Turki
- Amirah Al-Zahrani
- Alhanouf Al-haluli
- Sara Al-Abdulkarem
- Rawan Al Zayed
- Renad Al Haqbani
- Nouf Al Humaidhi
- Jude Al Khalifah
- Nouf Al Hussaini
- Alwateen Al Balawi
- Rahaf Al Shabri
- Danah Al Halees
- Rema Al Mutawa
- Amirah Al Dakhilallah
- Maha Al Nahdi
- Ghaida Al Braithen

THANKS!



Contact us:

